

LECTURE-22

Wednesday
22/4/26

ER to Relational Mapping.

- Adding weak entity as table: Since the primary key exist in parent.
∴ we'll add the primary key from parent entity into the weak entity's table.

Eg: "Book" is parent of "chapter". Book table will have 'BID' and 'Title' attributes and chapter table will have 'CID', 'Title' as well as 'BID' from parent i.e.: Book - Key of 'chapter' will be CID+BID.

- Foreign Key Approach:

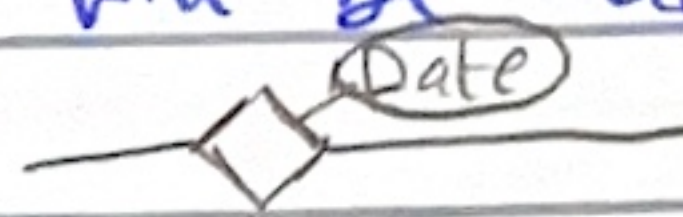


To join Student and Locker we've three option for foreign key:

- 1 - Adding Student's key in Locker table → Causes a lot of NULL problems
- 2 - Making a new table with key of student & Locker → Requires JOIN
- 3 - Adding Locker's key in student table → Perfect for this example.

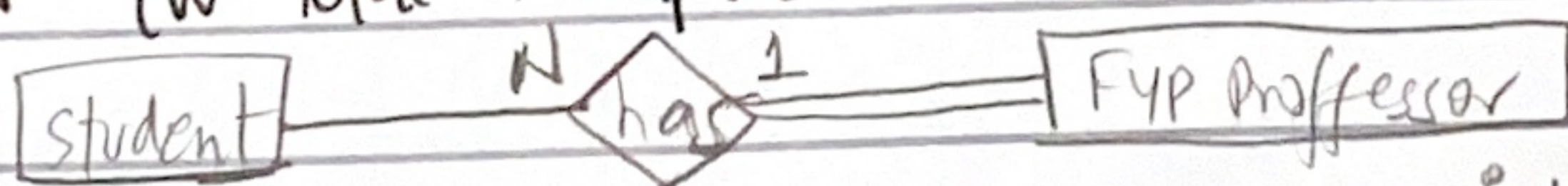
Rule: Jiske "total participation" hogi (double line); uski table main dosray ki key ayegi

NOTE: ∵ Since we used approach '3' in foreign key, so the attributes of relationship (diamond in diagram) will be added as column in Student table (w Locker's key).



⇒ Rule for Foreign Key adding in which table

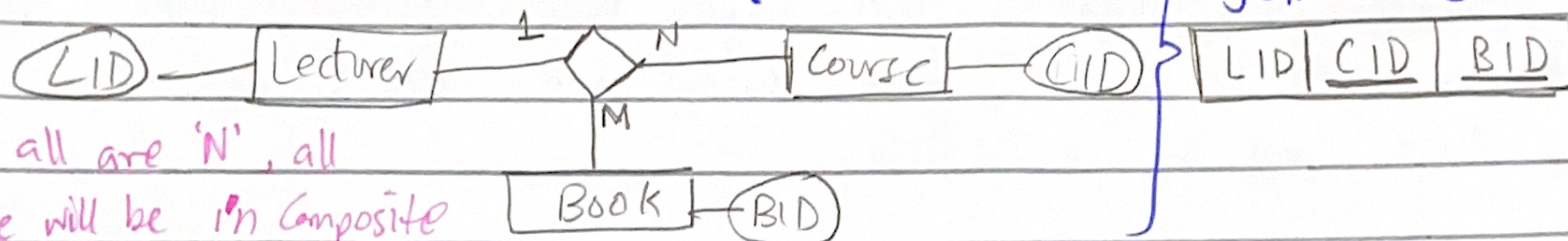
- 1:1 (w Total Participation) → Option 1 OR 2 depending which side of diamond has total participation.
- 1:1 (w No Total Participation) → Option 3: Make a JOIN table
- 1:N (w Total Participation on 'N' side) → Option 1/2 on N side's table
- 1:N (w Total Participation on '1' side) → Option 3 to Join



Here student table having Professor's key will NULLs since a lot of student's won't have an FYP Proff. Secondly having student's key in Proff. table will cause a lot redundancy.

- 1: N (w No Total Participation) $\rightarrow$ Mostly option 3: JOIN until some restraints in Company's schema ~~make~~ ^{makes} very less NULL or redundancy.
e.g: only CEO won't be employee so we can use option 1 or 2 which is better to have one NULL than having a JOIN in option 3.

- N: M $\rightarrow$ Always make JOIN table (option 3) . so NO REDUNDANCY.
- Relation with degree $> 2 \rightarrow$ Always make JOIN (option 3) $\uparrow$
Here the composite key in new table will be keys from each table which has 'N' cardinality. Since CID & BID is enough & LID will be unique.



* If all are 'N', all three will be in Composite Key

NOTE: Whenever we've a composite attribute like: address with extension: Street, City, House, then we'll only add street, city, house in the table NOT the address itself.